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2023-3R/2024 (MAT3204C)

2024

MATHEMATICS

Paper : MAT3204C

(Real Analysis)

Full Marks : 60

Time : Three hours

**The figures in the margin indicate
full marks for the questions.**

1. Answer the following questions : $2 \times 5 = 10$

(a) What is the Trichotomy property of R ?
If a, b, c are elements of R satisfying
 $a > b$ and $b > c$, then show that $a > c$.

~~(b)~~ Define a Cantor set.

~~(c)~~ Show that $\{1 + (-1)^n\}$ is not a Cauchy
sequence.

(d) Is the series $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots$ convergent?
Justify.

(e) Give example of a bounded sequence that
is not a Cauchy sequence.

Contd.

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2. Answer the following questions : $5 \times 4 = 20$

(a) Establish the following results :

- (i) If z and a are elements in R with $z + a = a$, then $z = 0$.
- (ii) If u and $b \neq 0$ are elements in R with $u \cdot b = b$, then $u = 1$.
- (iii) If $a \in R$ then $a \cdot 0 = 0$.

Or

Show that the set

$$A = \left\{ 1, -1, 1\frac{1}{2}, -1\frac{1}{2}, 1\frac{1}{3}, -1\frac{1}{3}, \dots \right\}$$

is closed but not open. Find interior and closure of A .

(b) State and prove the Bolzano-Weierstrass theorem for sets.

(c) Let the function f be defined by

$$f(x) = (2x^2 + 3x + 1)/(2x - 1) \text{ for}$$

$2 \leq x \leq 3$. Find a constant M such that

$$|f(x)| \leq M \text{ for all } x \text{ satisfying } 2 \leq x \leq 3.$$

(d) Show that $\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}$.

Further show that an arbitrary union of closed sets may not be closed.

Or

✓ State the Cauchy's criterion for convergence of an infinite series. Show

that the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

3. Answer the following questions : $10 \times 3 = 30$

(a) Let $\{x_n\}$ be a Cauchy sequence such that x_n is an integer for all $n \in \mathbb{N}$. Show that $\{x_n\}$ is a constant sequence. Moreover consider $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$ for $n \in \mathbb{N}$. Show that $\{x_n\}$ converges and find the limit.

$5+5=10$

Or

State and prove the monotonic convergence theorem. Show that the

sequence $\left\{ \left(1 + \frac{1}{n}\right)^n \right\}_{n=1}^{\infty}$ is convergent.

$5+5=10$

(b) State and prove Archimedean property. Applying it find the infimum of

$S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$. If $t > 0$, show that there

exists $n_t \in \mathbb{N}$ such that $0 < \frac{1}{n_t} < t$.

$5+2+3=10$

Or

$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$

What do you mean by supremum of a non-empty subset S of $\mathbb{R}$? If

$$S = \left\{ 1 - \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}, \text{ find } \sup S.$$

Again if $f(x) \leq g(y)$ for all $x, y \in D$, then show that $\sup f(D) \leq \inf g(D)$ where f, g are real-valued functions with common domain $D \subseteq \mathbb{R}$. Further state and prove Nested Intervals property.

$$1+2+2+5=10$$

(c) Test the convergency of the following series : 5+5=10

(i) $1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$

(ii) $\sum_{n=1}^{\infty} \frac{n^2 - 1}{n^2 + 1} x^n, x > 0$

Or

(i) Define conditional convergence of a series. Show that the series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \text{ is conditionally convergent.}$$

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(ii) Show that a sequence of real numbers is convergent if and only if it is Cauchy.

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